ELSEVIER Journal of Accounting and Economics 25 (1998) 283–319

JOURNAL OF Accounting & Economics

Accounting valuation, market expectation, and cross-sectional stock returns

Richard Frankel[a], Charles M.C. Lee[b,*]

[a] *School of Business Administration, University of Michigan, Ann Arbor, MI 48109-1234, USA*
[b] *Johnson Graduate School of Management, Cornell University, Ithaca, NY 14853-4201, USA*

Received 1 May 1997; accepted 7 August 1998

Abstract

This study examines the usefulness of an analyst-based valuation model in predicting cross-sectional stock returns. We estimate firms' fundamental values (V) using I/B/E/S consensus forecasts and a residual income model. We find that V is highly correlated with contemporaneous stock price, and that the V/P ratio is a good predictor of long-term cross-sectional returns. This effect is not explained by a firm's market beta, B/P ratio, or total market capitalization. In addition, we find errors in consensus analyst earnings forecasts are predictable, and that the predictive power of V/P can be improved by incorporating these errors. © 1998 Elsevier Science B.V. All rights reserved.

JEL classification: D4; G12; G14; M4

Keywords: Capital markets; Market expectations; Market efficiency; Valuation; Analyst forecasts

1. Introduction

Recent studies by Ohlson on residual income valuation have led empirical researchers to reexamine the relation between accounting numbers and firm value.[1] In this study, we operationalize the residual income model using analyst

* Corresponding author. Tel.: (607) 255-6255; fax: (607) 254-4590; e-mail: CL86@cornell.edu.

[1] Examples of Ohlson's work include Ohlson (1990, 1991, 1995) and Feltham and Ohlson (1995). Examples of empirical research include Bernard (1994), Fairfield (1994), Ou and Penman (1994), Penman and Sougiannas (1996), Abarbanell and Bernard (1995), and Frankel and Lee (1998), Lee et al. (1998), and Dechow et al. (1997).

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PII: S0165-4101(98)00026-3

earnings forecasts and examine its usefulness in predicting cross-sectional stock returns in the U.S. Specifically, we use I/B/E/S consensus earnings forecasts to proxy for market expectations of future earnings. We then use the resulting estimate of firm fundamental value (V_f) to investigate issues related to market efficiency and the predictability of cross-sectional stock returns.

We find that V_f estimates based on I/B/E/S consensus forecasts are highly correlated with contemporaneous stock prices. In recent years, V_f explains more than 70% of the cross-sectional variation in stock prices. Moreover, the value-to-price ratio (V_f/P) is a good predictor of cross-sectional returns, particularly over longer time horizons. In 12-month horizons, the V_f/P ratio predicts cross-sectional returns as well as the book-to-market ratio (B/P). However, over two or three year periods, buy-and-hold returns from V_f/P strategies are more than twice those from B/P strategies. Specifically, we find that higher V_f/P firms tend to earn higher long-term returns. This result is not due to differences in market betas, firm size, or the B/P ratio.

Because of its importance in estimating V_f, we also investigate the reliability of long-term I/B/E/S consensus earnings forecasts. We find that cross-sectional errors in the three-year-ahead consensus forecast are predictable. Specifically, we find some evidence that analysts tend to be more overly-optimistic in firms with higher past sales growth (*SG*) and higher P/B ratios. In addition, we find stronger evidence of over-optimism in firms with higher forecasted earnings growth (*Ltg*) and higher forecasted ROEs relative to current ROEs (*OP*). Combining these variables in a prediction model, we develop an estimate of the prediction error in long-term forecasts (*PErr*), and show this estimate has predictive power for cross-sectional returns.

Finally, we show the predictive power of PErr is incremental to a V_f/P strategy. During our sample period (1979–1991), a zero-cash investment strategy involving firms that are simultaneously in the top quintile of V_f/P and the bottom quintile of PErr yields cumulative buy-and-hold returns of more than 45% over 36 months. The three-year buy-and-hold strategy results in positive returns in both up and down markets. This effect is not explained by market beta, firm size, or the B/P ratio.

Our results contribute to the emerging literature on the residual income model in several ways. First, our analyst-based approach complements Penman and Sougiannas (1998), which uses ex post reported earnings. Second, we provide evidence on the reliability of I/B/E/S consensus forecasts for valuation, as well as a method for correcting predictable forecast errors. To our knowledge, this is the first study to develop a prediction model for long-run analyst forecast errors, and to trade profitably on that prediction.[2] Finally, we show that returns

[2] Several studies show analyst forecast errors differ for firms with certain characteristics, suggesting a relation between analyst forecast errors and various market pricing anomalies (e.g., Dechow and Sloan, 1997; Daniel and Mande, 1994; LaPorta, 1996; LaPorta et al., 1997). Other studies show

to a V_f/P strategy are not due to standard risk proxies, and that the strategy can be further improved by incorporating analyst forecast errors.

Our findings are also related to the finance literature on the predictability of stock returns. Much recent research has focused on accounting-based ratios that exhibit predictive power for stock returns. The B/P ratio, in particular, has been elevated to celebrity status by studies such as Fama and French (1992). Fama and French suggest B/P is a proxy for a firm's distress risk. However, little progress has been made in identifying the exact nature of this risk. Our results suggest that rather than attempting to produce a better risk proxy, superior return prediction may result from adopting a more complete valuation approach.

In sum, empirical studies involving equity valuation encounter two potential problems: (1) the use of overly restrictive models of intrinsic value, and (2) the use of biased proxies as model imputs. Our research design features a more robust valuation model than simple market-multiples, as well as a technique for improving on analysts' earnings forecasts. Our empirical findings suggest both will lead to better predictions of cross-sectional stock returns.

The remainder of this paper is organized as follows. In the next section, we present the accounting-based valuation model and describe its most salient features. In Section 3, we discuss the estimation procedures used to implement this model. Section 4 contains a discussion of the data and sample description. Section 5 reports the empirical results, and Section 6 concludes with a summary of our findings and their implications.

2. The residual income model

The valuation method we use in this study is a discounted residual income approach sometimes referred to as the Edwards–Bell–Ohlson (EBO) valuation technique.[3] Independent derivations of this valuation model have surfaced periodically throughout the accounting, finance and economics literature since the 1930s. In this section, we present the basic residual income equation and briefly develop the intuition behind the model.

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analysts may not use all available information when formulating their forecasts (e.g., Abarbanell, 1991; Abarbanell and Bernard, 1992; Stober, 1992). However, none of these studies develop a prediction model for analyst errors. Brown et al. (1995) do develop a prediction model for analyst errors, but their investment horizon is only one-quarter-ahead and the details of their model are proprietary.

[3] The term Edwards–Bell–Ohlson, or EBO, was coined by Bernard (1994). Theoretical development of this valuation method is found in Ohlson (1990, 1995), Lehman (1993), and Feltham and Ohlson (1995). Earlier treatments can be found in Preinreich (1938), Edwards and Bell (1961), and Peasnell (1982). For a simple guide to implementing this technique, see Lee (1996).

A stock's fundamental value is typically defined as the present value of its expected future dividends based on all currently available information. Notationally,

$$V_t^* \equiv \sum_{i=1}^{\infty} \frac{\mathrm{E}_t(D_{t+i})}{(1+r_\mathrm{e})^i}. \tag{1}$$

In this definition, V_t^* is the stock's fundamental value at time t, $\mathrm{E}_t(D_{t+i})$ is the expected future dividends for period $t+i$ conditional on information available at time t, and r_e is the cost of equity capital based on the information set at time t. This definition assumes a flat term-structure of discount rates.

It is easy to show that, as long as a firm's earnings and book value are forecasted in a manner consistent with clean surplus accounting, Eq. (1) can be rewritten as the reported book value, plus an infinite sum of discounted residual income:

$$\begin{aligned} V_t^* &= B_t + \sum_{i=1}^{\infty} \frac{\mathrm{E}_t[NI_{t+i} - (r_\mathrm{e}\, B_{t+i-1})]}{(1+r_\mathrm{e})^i} \\ &= B_t + \sum_{i=1}^{\infty} \frac{\mathrm{E}_t[(ROE_{t+i} - r_\mathrm{e})\, B_{t+i-1}]}{(1+r_\mathrm{e})^i}, \end{aligned} \tag{2}$$

where B_t is the book value at time t, $E_t[.]$ is expectation based on information available at time t, NI_{t+i} is the Net Income for period $t+i$, r_e is the cost of equity capital and ROE_{t+i} is the after-tax return on book equity for period $t+i$.

Note that this equation is identical to a dividend discount model, but expresses firm value in terms of accounting numbers. It therefore relies on the same theory and is subject to the same theoretical limitations as the dividend discount model. However, the model provides a framework for analyzing the relation between accounting numbers and firm value.

Eq. (2) shows that equity value can be split into two components – an accounting measure of the capital invested (B_t), and a measure of the present value of future residual income, defined as present value of future discounted cash flows not captured by the current book value. If a firm earns future accounting income at a rate exactly equal to its cost of equity capital, then the present value of future residual income is zero, and $V_t = B_t$. In other words, firms that neither create nor destroy wealth relative to their accounting-based shareholders' equity, will be worth only their current book value. However, firms whose expected ROEs are higher (lower) than r_e will have values greater (lesser) than their book values.

If the market price approximates future discounted cash flows, then Eq. (2) offers a natural interpretation for the price-to-book ratio. Dividing both sides of Eq. (2) by B_t, we can express P/B in terms of a firm's future abnormal ROEs. In a competitive equilibrium, a typical firm's ROE should be close to its

cost of equity capital ($ROE = r_e$), and the typical price-to-book ratio should be close to 1.[4] Moreover, firms expected to earn above (below) normal ROEs in the future should trade at higher (lower) price-to-book ratios. Stated another way, this equation properly impounds a firm's expected future profitability into its equity value estimate.

Eq. (2) shows that future earnings performance should be closely linked to current B/P ratios. This association is consistent with the empirical finding that low (high) B/P firms have higher (lower) future ROEs (e.g., FF, 1995; Fairfield, 1994; Bernard, 1994). However, the inverse relation between future earnings performance and current B/P ratios should not be interpreted as proof of market efficiency – it shows that the market *considers* future profitability when formulating prices, not that it *fully incorporates* all available information when doing so. Later, we use analyst forecasts of future earnings to directly examine whether the market fully incorporates current information in establishing prices.

Current literature shows that a number of ad hoc variables such as cash flow yield (Chan et al., 1991; LSV, 1994; Davis, 1994), earnings yield (Basu, 1977; Jaffe et al., 1989) and dividend yield (Litzenberger and Ramaswamy, 1979) have predictive power for cross-sectional returns. These yield measures have often been interpreted as risk proxies. Eq. (2) suggests that these market multiples may also work because their accounting component reflects (imperfectly) some dimension of V_t^*. However, neither book value nor earnings, is sufficient to capture V_t^* . One of the objectives in this paper is to use the EBO model to derive a more precise estimate of V_t^*, and examine whether a more complete valuation model yields superior power to predict risk adjusted returns.

Several recent studies evaluate this model's ability to explain stock prices. Penman and Sougiannas (1998) implement variations of the model using ex post realizations of earnings to proxy for ex ante expectations. Lee et al. (1998) operationalize the model for the 30 stocks in the Dow Jones Industrial Average and examine time-series properties of the model. Frankel and Lee (1998) employ the model in an international context and find that V has high explanatory power for prices in 21 countries. More recently, both Francis et al. (1997) and Dechow et al. (1997) examine the empirical properties of the model under alternative specifications. Except for Dechow et al. (1997), these studies do not examine the predictive power of the model for cross-sectional stock returns in the US.

[4] In practice, firms' reported ROE may differ from their costs of equity in competitive equilibrium due to accounting and risk factors. For example, we show later (in Table 1) that average P/Bs are above 1, and average ROEs are somewhat above firms' costs of capital. These observations are consistent with the fact that accounting systems that are, on average, conservative.

3. Model estimation procedures

Eq. (2) presents a simple procedure for estimating a firm's intrinsic value (V_t). The four parameters needed for the estimation are: the cost of equity capital (r_e), future ROE forecasts (FROEs), current book value (B_t), and a dividend payout ratio (k). The first three parameters' roles are readily seen in Eq. (2). The last input, the dividend payout ratio (k), is used in conjunction with the clean surplus relation (CSR) to derive future book values. In this section, we discuss the specifics of the model estimation procedure.

Cost of equity capital (r_e). In theory, r_e should be firm-specific, reflecting the premium demanded by equity investors to invest in a firm or project of comparable risk. In practice, however, there is little consensus on how this discount rate should be determined. For this study, we use three different approaches – a constant discount rate, and two industry-based discount rates derived by FF (1997). The FF discount rates are based on a one-factor and a three-factor risk model.

Dividend payout ratio (k). The dividend payout ratio is the percentage of net income paid out in the form of dividends each year. We obtain a firm-specific estimate of k by dividing the common stock dividends paid in the most recent year (Compustat Item 21) by net income before extraordinary items (Compustat Item 237). For firms with negative earnings (approximately 11% of our sample), we divide dividends by six percent of total assets to derive an estimated payout ratio.[5] This variable is used, in conjunction B_t, to derive forecasted book values:

$$\begin{aligned} B_{t+1} &= B_t + NI_{t+1} - d_{t+1} = B_t + (1-k)NI_{t+1} \\ &= [1 + (1-k)ROE_{t+1}]B_t. \end{aligned}$$

Analogously, all future book values can be expressed as functions of B_t, k, and future ROEs. For example, we can write

$$B_{t+2} = [1 + (1-k)ROE_{t+1}][1 + (1-k)ROE_{t+2}]B_t.$$

Future ROEs. The most important and difficult task in the EBO valuation exercise is forecasting future ROEs (or, equivalently, forecasting future earnings). Two alternatives, based on ex ante information, are: (1) use prior period earnings (or ROEs), or (2) use analysts' earnings forecasts (e.g., Abarbanell and Bernard (1995) use earnings forecasts from Value-Line]. We use both methods and derive a value metric based on historical earnings (V_h), as well as a value metric based on consensus I/B/E/S analyst forecasts (V_f).

[5] Six percent reflects the average long-run return-on-assets (see Table 1, last column). We use this measure as a proxy for normal earnings when reported earnings are negative. As explained later, we also constrain k to be between 0 and 100%.

Fairfield et al. (1994) show that, in large samples, the correlation between current year ROEs and next year's ROEs is around 0.66, suggesting that the current period ROE is a reasonable starting point for estimating future ROEs. The use of I/B/E/S data should result in a more precise proxy for market expectations of earnings. Prior studies show that analyst earnings forecasts are superior to time-series forecasts (e.g., O'Brien, 1988; Brown et al., 1987a,b). However, the predictive superiority of an analyst-based value metric (V_{f}) over a historical-based value metric (V_{h}) is an open empirical question.[6]

Forecast horizons and terminal value estimation. Eq. (2) expresses firm value in terms of an infinite series, but for practical purposes, the explicit forecast period must be finite. This limitation necessitates a terminal value estimate – that is, an estimate of the value of the firm based on residual income earned after the explicit forecasting period. One approach is to estimate the terminal value by first expanding Eq. (2) to T terms, and then taking the next term in the expansion as a perpetuity. For example, if the explicit forecast period ends after T periods, the terminal value is:

$$\frac{(ROE_{T+1} - r_{\mathrm{e}})}{(1 + r_{\mathrm{e}})^T r_{\mathrm{e}}} B_T.$$

This procedure is mathematically equivalent to a T-period discounted dividend model in which year $T + 1$ earnings is treated as a perpetuity (see Penman, 1995). The resulting value estimate therefore depends critically on the particular earnings forecast used in the terminal value. Various alternative approaches have appeared in the literature. For example, both Lee et al. (1998) and Dechow et al. (1997) feature various permutations for the terminal value.

In this study, we take a simple approach using a short-horizon earnings forecasts of up to three years.[7] In theory, T should be set large enough for firms to reach their competitive equilibrium. However, our ability to forecast future ROEs diminishes quickly over time, and forecasting errors are compounded in longer expansions. Therefore, we estimate three forms of V_t:

$$\hat{V}_t^1 = B_t + \frac{(FROE_t - r_{\mathrm{e}})}{(1 + r_{\mathrm{e}})} B_t + \frac{(FROE_t - r_{\mathrm{e}})}{(1 + r_{\mathrm{e}}) r_{\mathrm{e}}} B_t, \tag{3.1}$$

[6] Abarbanell and Bernard (1995) use Value-Line forecasts to estimate a similar EBO valuation equation in addressing the question of whether U.S. markets are myopic. Under the null hypothesis of market efficiency, they examine whether markets underprice long-run earnings relative to near-term earnings. They also provide evidence that future forecasted ROEs may not be fully impounded in current prices.

[7] We also estimated a 12-period expansion of the formula in which ROEs are reverted back to the industry median. The 12-period version had slightly lower correlation with stock prices and similar predictive power for returns.

$$\hat{V}_t^2 = B_t + \frac{(FROE_t - r_e)}{(1 + r_e)} B_t + \frac{(FROE_{t+1} - r_e)}{(1 + r_e) r_e} B_{t+1}, \tag{3.2}$$

$$\hat{V}_t^3 = B_t + \frac{(FROE_t - r_e)}{(1 + r_e)} B_t + \frac{(FROE_{t+1} - r_e)}{(1 + r_e)^2} B_{t+1} + \frac{(FROE_{t+2} - r_e)}{(1 + r_e)^2 r_e} B_{t+2}. \tag{3.3}$$

Eq. (3.1) represents a two-period expansion of the residual income model with the forecasted ROE for the current year ($FROE_t$) assumed to be earned in perpetuity. Eq. (3.2) also represents a two-period expansion of the model, but we use a two-year-ahead forecasted ROE ($FROE_{t+1}$)in the perpetuity. Similarly, Eq. (3.3) is a three-period model.

The right-hand side of each equation consists of ex ante observables. To estimate V_h, we use the return on average equity, $ROE_t = NI_t/[(B_t + B_{t-1})/2]$, to proxy for all future ROEs – i.e., we substitute ROE_t for all the $FROE$s in the above equations.[8] NI_t is earnings to common shareholders in year t, net of extraordinary items, taxes, and preferred dividends (Compustat Item 237), and B_t is total common shareholders' equity from year t (Compustat Item 60). To estimate V_f, we derive future ROEs and book values from I/B/E/S consensus forecasts using a sequential procedure described in the Appendix.

4. Data and sample description

The original sample of firms consists of all domestic nonfinancial companies in the intersection of (a) the NYSE, AMEX, and NASDAQ return files from the Center for Research in Security Prices (CRSP) and (b) a merged Compustat annual industrial file, including PST, full coverage and research files. We require firms to meet the Compustat data requirements (for B_{t-1}, B_{t-2}, NI_{t-1}, and DIV_{t-1}) and have the necessary CRSP stock prices and shares outstanding data (for fiscal year end $t-1$, and the end of June in year t). Furthermore, we require firms to have a one-year-ahead and a two-years-ahead earnings-per-share (EPS) forecast from I/B/E/S. Because I/B/E/S began operations in 1975, this requirement limits our sample period to 1975–93. We further constrain our sample to firms with fiscal-year-ends between June and December, inclusively. Because we use I/B/E/S forecasts issued in May, this constraint ensures that forecasted earnings correspond to the correct fiscal year. Using V_h for the entire Fama and French (1992) sample over the period of 1962–1993 yields similar results. The results are also similar for a sample consisting of just December year-end firms.

[8] In theory, the model calls for beginning-of-year book values. However, we use the annual average to avoid situations where an unusually low book value in year t-1 inflates forecasted ROEs.

To ensure that accounting variables are known before returns are computed, we allow a minimum gap of six months between the fiscal-year-end and the portfolio formation date. Specifically, we match accounting data for all fiscal year ends in the calendar year $t-1$ to returns on portfolios formed at the end of June of year t. We use a firm's market equity at its fiscal-year-end to compute its book-to-market and value-to-market ratios, and the market equity on June 30 of year t to measure its size. These procedures are similar to FF (1992), except their B/P ratios are based on December market prices, rather than fiscal year end. They report that this difference has little effect on their return tests.

In estimating V_{f}, we use the I/B/E/S mean (also called consensus) forecast from the May statistical period of year t. This mean estimate is determined from analyst forecasts on file with I/B/E/S as of the Thursday after the third Friday of each month.[9] Since these monthly reports are widely available soon after each computer run, the May statistics are in the public domain well before our portfolio formation date. Our valuation formula uses three pieces of I/B/E/S data: earnings-per-share forecasts one-year-ahead ($FY1$), EPS forecasts two-years-ahead ($FY2$), and a five-year long-term growth rate (Ltg). Between 1975 and 1979, analysts reported just $FY1$ and $FY2$, but after 1980, most firms also had Ltg information. The Appendix explains the procedure we followed to derive future ROE forecasts when all three variables are not in the May I/B/E/S report.

In estimating Eqs. (3.1), (3.2) and (3.3), we remove firms with negative book values, because ROEs for these firms cannot be interpreted in economic terms. In addition, some firms have extremely low book values, or earnings, leading to unreasonable ROE or k estimates. We eliminate such firms by considering only firms with ROEs or FROEs of less than 100% and dividend payout ratios of less than 100%. These procedures eliminate 1075 firm-years. We also remove 51 firms with stock prices of under \$1 as of the end of June in year t. These firms have unstable B/P, V_{h}/P and V_{f}/P ratios and poor market liquidity (that is, they cannot be included in equal-weighted portfolios without incurring disproportionally large trading costs).[10]

Taken together, our filters eliminated 1,126 observations (approximately 5%), leaving a final sample of 18,162 firm-years. These common sense filters ensure the subsequent results are not driven by outliers. Further, the strategies we examine are tradable, in the sense that all the portfolios are constructed using firm characteristics that are observable at the time of portfolio formation.

[9] Using median rather than mean forecasts is unlikely to affect results because the distribution of forecasted growth is quite symmetric.

[10] See Ball et al. (1995) for additional evidence on the sensitivity of contrarian returns to the removal of stocks with prices under \$1.

5. Empirical results

Table 1 reports annual summary statistics for the total sample. The average dividend payout ratio ranges from a high of 35% in 1976 and 1978 to a low of 19% in 1993. The average return-on-equity ranges between 8 and 18%. The average book value per share over the period is \$16.87. The average P/B ratio is 2.18; however, this ratio is inflated by the presence of low book value firms in the sample. Taking the inverse of the B/P ratio [shown as 1/Avg(B/P)] reduces the average P/B ratio to 1.43. Finally, Table 1 reports the return-on-asset ratio [Avg ROA], which averages 6%. Collectively, these results illustrate the stability of the key model parameters over our sample period.

Table 1
Summary statistics by year

Table values represent annual, equally-weighted average statistics for the sample firms. Year $t-1$ is the year from which the accounting data are obtained. ME is the market value of equity as of June 30 of year t in millions of dollars. k is the dividend payout ratio, computed as common stock dividends divided by earnings to common shareholders. For firms with negative earnings, k is computed as common stock dividends divided by (total assets $\times 0.06$). ROE is the return on equity for year $t-1$ computed as net income for year $t-1$ divided by the year $t-1$ average book equity. B is the year $t-1$ reported book value per share. P is the stock price as of June 30 in year t. ROA is the return on total assets for year $t-1$. 1/(Avg.B/P) is the inverse of the equally-weighted average B/P ratio for each year. Averages reported in the bottom row represent time-series means of the annual statistics.

Year t	No. firm	Avg. ME	Avg. k	Avg. ROE	Avg. B	Avg. P/B	1/(Avg. B/P)	Avg. ROA
76	361	1168	0.35	0.14	21.77	2.02	1.40	0.07
77	312	1264	0.32	0.16	22.83	1.69	1.30	0.08
78	535	863	0.35	0.16	22.69	1.52	1.15	0.08
79	675	740	0.32	0.17	22.51	1.52	1.13	0.08
80	718	875	0.33	0.18	22.26	1.59	1.05	0.08
81	812	921	0.32	0.16	20.81	1.96	1.27	0.07
82	920	693	0.32	0.15	20.44	1.38	0.94	0.07
83	952	1049	0.33	0.12	18.44	2.67	1.60	0.06
84	1174	781	0.27	0.11	15.66	1.99	1.34	0.05
85	1130	1002	0.25	0.13	15.74	2.11	1.44	0.06
86	1146	1269	0.25	0.10	14.73	2.69	1.71	0.05
87	1213	1387	0.23	0.09	12.97	2.82	1.81	0.04
88	1228	1282	0.22	0.11	12.83	2.34	1.61	0.05
89	1298	1423	0.21	0.13	13.17	2.33	1.64	0.06
90	1306	1506	0.22	0.12	12.65	2.51	1.53	0.06
91	1352	1591	0.23	0.11	12.57	2.54	1.45	0.05
92	1423	1579	0.22	0.08	11.36	2.63	1.54	0.04
93	1607	1605	0.19	0.09	10.17	2.96	1.82	0.04
All years	18162	1167	0.27	0.13	16.87	2.18	1.43	0.06

5.1. Correlation with stock prices

Table 2 reports cross-sectional Spearman rank correlation coefficients between stock prices and either book value (*B*) or one of six EBO value metrics. These six measures reflect the three different empirical estimates of value discussed earlier (Eqs. (3.1), (3.2) and (3.3)), estimated using historical earnings and analyst forecasts. The discount rates used are industry specific cost-of-equity

Table 2
Annual cross-sectional correlation of stock prices to book equity and EBO value measures

Table values represent cross-sectional Spearman correlation coefficients between the stock price on June 30 of year *t*, book equity per share in calendar year $t-1$ (*B*), and two sets of three fundamental value metrics computed using the Edwards–Bell–Ohlson (EBO) formula. The historical *EBO* value measures use current year return-on-equity (*ROE*) to proxy for future ROEs. The analyst based *EBO* value measures use consensus I/B/E/S forecasts to proxy for future ROEs. Each set of nine value measures differs only in the assumed the number of forecasting periods ($T = 1$, 2, or 3). The discount rate used is a three-factor industry-specific cost-of-equity (Fama and French, 1997). All years represents the time-series mean of annual cross-sectional correlations.

			FF Three-factor			FF Three-factor		
			$T=1$	$T=2$	$T=3$	$T=1$	$T=2$	$T=3$
Year *t*	*Obs.*	*B*	Historical *EBO* value measures			Analyst based *EBO* value measures		
76	361	0.48	0.68	0.68	0.68	0.73	0.73	0.74
77	312	0.56	0.73	0.72	0.70	0.78	0.79	0.78
78	535	0.49	0.71	0.72	0.72	0.79	0.79	0.79
79	675	0.54	0.68	0.68	0.68	0.76	0.77	0.77
80	718	0.43	0.65	0.67	0.68	0.76	0.79	0.80
81	812	0.45	0.64	0.65	0.66	0.70	0.72	0.74
82	920	0.56	0.73	0.73	0.72	0.79	0.82	0.82
83	952	0.45	0.54	0.53	0.53	0.66	0.70	0.72
84	1174	0.69	0.71	0.69	0.68	0.81	0.82	0.82
85	1130	0.72	0.83	0.82	0.81	0.88	0.89	0.89
86	1146	0.69	0.76	0.74	0.74	0.87	0.87	0.87
87	1213	0.70	0.70	0.68	0.68	0.82	0.85	0.85
88	1228	0.74	0.77	0.75	0.74	0.87	0.89	0.88
89	1298	0.76	0.78	0.77	0.76	0.87	0.88	0.87
90	1306	0.67	0.74	0.73	0.72	0.84	0.86	0.87
91	1352	0.63	0.73	0.71	0.71	0.82	0.86	0.86
92	1423	0.64	0.64	0.61	0.60	0.81	0.83	0.83
93	1607	0.63	0.64	0.61	0.60	0.77	0.80	0.80
All years	18162	0.60	0.70	0.69	0.69	0.80	0.81	0.82

based on a three-factor model (Fama and French, 1997).[11] The three-factor model is estimated using mimicking portfolios for firm size and market-to-book ratios. In estimating these models, firms are grouped into 48 industry classes. The three-factor industry cost-of-equity ranges between 8.23% (for Alcoholic Beverages) and 16.49% (for Real Estate).[12]

Over our sample period, book value (B) had an average correlation with price of 0.60, suggesting that book equity explains around 36% of the cross-sectional variation in prices. Compared to B, each of the six value measures displays a higher average correlation with stock price. V_h explains around 49% of the cross-sectional variation in prices. Increasing T from 1 to 3 produces slightly weaker correlations, probably due to rapid decay in the precision of the ROE forecasts over time. In sum, values based on historical ROEs contain important value-relevant information not captured by B.

Table 2 also shows that the cross-sectional correlation with price increases when EBO value is estimated using analyst forecasts. Over our sample period, V_f explains around two-thirds of the cross-sectional variation in prices. Increasing T from 1 to 3 produces slightly better correlations, but varying the discount rate again has little effect. Evidently, analysts' earnings forecasts contain more value-relevant information than is reflected in historical ROEs. The superiority of analysts over simple random-walk models in forecasting earnings is well documented (Fried and Givoly, 1982; Brown et al., 1987a,b; O'Brien, 1988). Our findings suggest analyst forecasts also better reflect the market expectations of earnings implicit in the EBO model.

Table 2 suggests that using all variations of the estimated V metric is unnecessary. Therefore, for the remainder of this paper, we use V_h to denote the fundamental value computed from historical ROEs, a three-factor industry-specific discount rate, and a forecast horizon of two-period (Eq. (3.2)). We use V_f to denote the fundamental value computed using the mean analyst forecast, a three factor industry-specific discount rate, and a forecast horizon of three periods (Eq. (3.3)).[13]

[11] We also used constant interest rates of 11%, 12%, and 13% as well as an industry-specific single-factor model. We find varying the discount rate had little effect on our results. Abarbanell and Bernard (1995) also find that allowing for intertemporal and firm-specific variations in r_e had little effect on their results.

[12] Specifically, we use the risk premium for each industry reported in Fama and French (1997), Table 7, plus a constant riskless rate of 0.0646 per year – the average annualized 30-day t-bill rate over our sample period. The risk premiums we used are computed from 5-year rolling regressions by industry. See Fama and French (1997) for more details.

[13] We used a two-period model for V_h because of concerns for the accuracy of the time-series earnings model beyond two years. However, using a three-period model for V_h yields similar results.

5.2. Correlation with future returns: uni-dimensional analyses

The main focus of this study is on the prediction of future returns. As a first step, we construct uni-dimensional portfolios based on market value of equity (ME), B/P, and V_f/P. Table 3 reports characteristics of quintile portfolios formed on the basis of these firm characteristics. This table is constructed by sorting all sample firms into quintiles at the end of June each year. Firm size quintiles are formed in two ways. First, as in Fama and French (1992), the size decile cutoffs are based on June 30 prices for all NYSE firms. Second, we use annual in-sample firm size cutoffs to form quintiles. For each portfolio, Table 3 reports the average B/P, ME, and V_f/P values, as well as the average post-ranking market betas, and average buy-and-hold return over the next 12 months ($Ret12$), 24 months ($Ret24$), and 36 months ($Ret36$). Market beta for each firm is estimated using an equal-weighted market index and each firm's monthly returns over the next 36 months. The last row in each panel shows the number of firm-year observations in each portfolio, and applies to all variables except the stock return variables. When we require availability of stock returns, the number of observations drop to 16,549, 14,385, and 12,377 for $Ret12$, $Ret24$ and $Ret36$, respectively.

The right column of Table 3 reports the differences in means between the top ($Q5$) and bottom ($Q1$) quintiles. The statistical significance of this difference is assessed using a Monte Carlo simulation technique similar to those discussed in Barber and Lyon (1997), Kothari and Warner (1997), and Lyon et al. (1998). Specifically, we form empirical reference distributions by randomly assigning the population of eligible firms each year into quintile portfolios (without replacement). This procedure generates five random quintile portfolios each year with the same number of observations as the actual quintile portfolios. We repeat the process until we have obtained 1000 sets of quintile portfolios for each year. We then compute the mean returns for the $Q5$–$Q1$ portfolio. To determine statistical significance, we use p-values calculated from the simulated empirical distribution of mean $Q5$–$Q1$ returns.

Our randomization procedure avoids the three main econometric problems discussed in Lyon et al. (1998) and Kothari and Warner (1997). First, our reference portfolios only contain firms that are available for investing at the same time as our sample firms. This avoids the new listing or survivor bias. Second, we compute returns for the reference portfolios in exactly the same manner as for the actual portfolios (that is, both reflect buy-and-hold returns over the same time horizon). This avoids the re-balancing bias, and adjusts for serial correlations in returns induced by overlapping holding periods. Finally, the use of p-values calculated from the simulated empirical distribution avoids the skewness bias discussed in the literature.

Table 3
Characteristics of quintile-portfolios formed by ME, B/P, and V_f/P

This table reports the characteristics of quintile portfolios formed at the end of June each year by market value of equity (ME), book-value-to-price (B/P), and analyst based EBO value-to-price (V_f/P). Each panel reports mean values for individual quintile characteristics. ME is the market value of shareholders' equity as of June 30 of year t, expressed in millions. ME quintiles are based on size cutoffs for all NYSE firms (Panel A) and on in-sample size cutoffs (Panel B). Book value (B) is book equity per share in calendar year $t-1$. Price (P) is the stock price at the end of June in year t. Analyst based EBO value (V_f) is a fundamental value measure derived using current I/B/E/S consensus analyst predictions of future earnings available prior to June 30 of year t. beta is estimated using monthly returns over the 36 months beginning July of year t. $Ret12$, $Ret24$, and $Ret36$ are the average one-year, two-year, three-year buy-and-hold return for the portfolio. Obs. is the number of observations in each quintile and applies to all variables except $Ret12$, $Ret24$, and $Ret36$. Results in the All Firms column represent unconditional means. $Q5-Q1$ Diff. results represent differences in means between the top ($Q5$) and bottom ($Q1$) quintiles. The statistical significance of this difference is derived using Monte Carlo simulation. Specifically, we form empirical reference distributions by randomly assigning eligible firms into quintiles each year. ***, **, * signify that the observed difference between the extreme quintiles is significantly different from those of the reference distribution at the 1%, 5% and 10% levels, respectively (two-tailed). The sample period is 1977–1992 ($t = 77$ to 92).

Panel A – Market-equity portfolios (NYSE size quintiles)

	$Q1$ (Low ME)	$Q2$	$Q3$	$Q4$	$Q5$ (High ME)	All Firms	$Q5-Q1$ Diff.
ME	9	25	59	157	2293	1230	
B/P	1.39	1.04	0.79	0.66	0.62	0.69	−0.77
V_f/P	1.33	1.04	0.92	0.87	0.91	0.91	−0.42
Beta	0.73	1.06	1.14	1.17	1.03	1.08	0.30
$Ret12$	0.379	0.239	0.153	0.159	0.146	0.159	−0.233***
$Ret24$	0.515	0.340	0.304	0.312	0.305	0.311	−0.210*
$Ret36$	0.835	0.556	0.442	0.487	0.497	0.493	−0.338*
Obs.	232	1144	2692	4761	9333	18162	

Panel B – Market-equity portfolios (in-sample size quintiles)

	$Q1$ (Low ME)	$Q2$	$Q3$	$Q4$	$Q5$ (High ME)	All Firms	$Q5-Q1$ Diff.
ME	41	117	277	722	4983	1230	
B/P	0.92	0.69	0.61	0.65	0.60	0.69	−0.32
V_f/P	1.00	0.90	0.85	0.87	0.95	0.91	−0.05
Beta	1.08	1.17	1.11	1.07	0.98	1.08	−0.10
$Ret12$	0.158	0.157	0.161	0.158	0.159	0.159	0.001
$Ret24$	0.285	0.311	0.319	0.310	0.330	0.311	0.045**
$Ret36$	0.459	0.486	0.489	0.503	0.525	0.493	0.066**
Obs	3622	3636	3632	3632	3640	18162	

Table 3 (continued)

Panel C – Book-to-price (B/P) Portfolios	Q1 (Low B/P)	$Q2$	$Q3$	$Q4$	$Q5$ (High B/P)	All Firms	$Q5-Q1$ Diff.
B/P	0.24	0.42	0.60	0.81	1.39	0.69	–
ME	1641	1434	1345	1068	666	1230	−975
V_f/P	0.75	0.86	0.93	1.02	1.01	0.91	0.260
Beta	1.29	1.17	1.05	0.93	0.97	1.08	−0.320
$Ret12$	0.137	0.148	0.156	0.166	0.186	0.159	0.049***
$Ret24$	0.251	0.300	0.332	0.338	0.333	0.311	0.082***
$Ret36$	0.407	0.450	0.513	0.535	0.558	0.493	0.151***
Obs	3621	3628	3634	3636	3643	18162	

Panel D – V_f/P portfolios	$Q1$ (Low V_f/P)	$Q2$	$Q3$	$Q4$	$Q5$ (High V_f/P)	All Firms	$Q5-Q1$ Diff.
V_f/P	0.40	0.70	0.87	1.06	1.54	0.91	–
B/P	0.60	0.59	0.68	0.75	0.85	0.69	0.25
ME	812	1252	1531	1377	1177	1230	365
Beta	1.24	1.09	1.05	0.99	1.03	1.08	−0.210
$Ret12$	0.138	0.154	0.159	0.172	0.169	0.159	0.031***
$Ret24$	0.217	0.298	0.317	0.351	0.369	0.311	0.152***
$Ret36$	0.331	0.450	0.491	0.549	0.637	0.493	0.306***
Obs	3626	3632	3632	3632	3640	18162	

Despite these advantages, this simulation procedure still has a potential deficiency.[14] By randomly assigning firms to quintiles 1 through 5, the procedure creates portfolios whose covariance is equal to the average covariance across all returns in our sample. If the actual correlation structure between portfolios one and five differs significantly from this average covariance, the p-values based on the simulation may be misleading. In particular, cross-correlations in the data may cause the variance of the simulated $Q5$–$Q1$ returns to be lower than that of the true $Q5$–$Q1$ returns. To mitigate this problem, we will provide additional corroborating evidence using more traditional statistical procedures (see Tables 8 and 9).

Panels A and B examine the ME effect for our sample. Panel A shows that a small-firm effect exists when NYSE size quintiles are used. Over 12, 24, and 36 month periods following portfolio formation, small firms generally outperform large firms. However, because we require that firms be followed by analysts, larger firms dominate the sample – the last row of Panel A shows that over 80% of our firms are larger than the median NYSE firm. Panel B shows

[14] We thank the referee for his insights on this point.

that when our firms are grouped by in-sample size cutoffs, large firms actually outperform small firms over 24 and 36 month holding periods. Thus, in our sample, the firm-size effect is driven primarily by a small set of the smallest stocks.

Panel C confirms a B/P effect for our sample. The lowest B/P quintile firms earn an average return of 13.7% over the next 12-months while the highest B/P quintile firms earn 18.6%. The difference of 4.9% is statistically significant at 1%, and is comparable in magnitude to other studies reporting the B/P effect using I/B/E/S-constrained samples (e.g., Dechow and Sloan, 1997). The B/P effect is also seen over longer holding periods. The relation between B/P and future returns is generally monotonic across the quintiles. As in Lakonishok et al. (1994), we find low B/P firms have higher betas than high B/P firms. This result suggests that the B/P effect is not due to differences in market risk. High B/P betas may also be biased downward due to nonsychroneity, although the larger nature of our sample firms may reduce this problem.[15]

Panel D shows that V_f/P portfolios have some similarities with B/P portfolios. V_f/P and B/P are positively correlated. High V_f/P firms, like high B/P firms, tend to have lower market betas. Moreover, these results show that V_f/P also predicts returns. The short-term prediction results for V_f/P are slightly weaker than the results for B/P. The lowest V_f/P quintile firms earn 13.8% over the next 12 months, while the highest V_f/P quintile firms earn only 16.9%. Furthermore, the returns pattern is not monotonic across the quintiles. However, over 24 and 36 months, high V_f/P firms significantly outperform low V_f/P firms. Indeed, over these longer horizon, we observe a monotonic relation in returns across the quintiles. Over 36 months, for example, the spread between the highest and lowest V_f/P portfolios is 30.6%.

Fig. 1 illustrates the cumulative returns from a 36 month buy-and-hold strategy involving B/P and V_f/P. To construct this graph, a long-position is taken in the top quintile firms based on each ratio, and a short-position is taken in the bottom quintile firms. This graph reports the difference in cumulative buy-and-hold returns between the top and bottom quintiles at monthly intervals over the next three years. The graph shows that over a 36 month period, the V/P strategy outperforms the B/P strategy by a wide margin.[16]

In summary, we find that V_f/P is much better at explaining cross-sectional prices than B/P. V_f/P is also a better predictor of long-term returns. However, V_f/P is not necessarily more useful for predicting returns over 12 month intervals. Our findings suggest that while analyst consensus forecasts provide

[15] Kothari et al. (1995) show annual betas are more highly correlated with cross-sectional returns.

[16] Results for V_h/P are similar to those for V_f/P. Interestingly, returns from the B/P strategy exhibit a seasonal pattern not found in the other two strategies – Fig. 1 shows that the January effect is much more pronounced in the B/P strategy than in the V/P strategy.

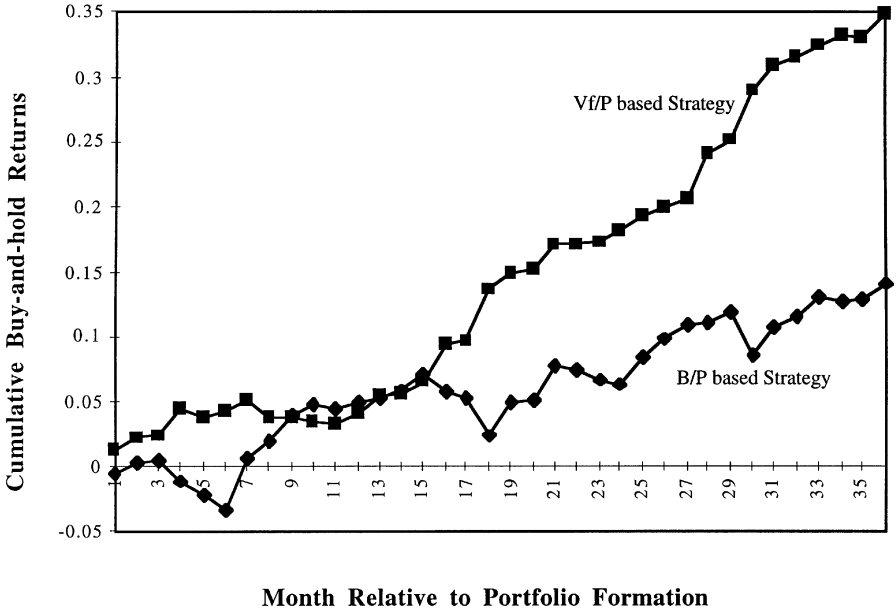

Fig. 1. Cumulative (buy-and-hold) returns produced by B/P and V_f/P Trading strategies. This figure shows the cumulative (buy-and-hold) returns from B/P and V_f/P based trading strategies. B is book equity per share in calendar year $t-1$. P is price per share at the end of June 30 of year t. V_f is a fundamental value estimate based on the consensus analyst forecast as of May of year t. Each year, portfolios are formed at the end of June by sorting firms into quintiles on the basis of B/P and V_f/P. For each investment strategy, this graph depicts the cumulative buy-and-hold returns produced by buying firms in the top quintile and selling firms in the bottom quintile at the beginning of July, and maintaining these investments until the end of the indicated month. The sample period is 1979–1991 (year $t = 1979$ to 1991).

a good proxy for market expectations, trading on the basis of these forecasts does not necessarily yield higher short-run (12 month) returns than trading on B/P. In later tests, we explore a strategy that seeks to improve on the consensus earnings forecast.

5.3. Correlation with future returns: bi-dimensional analyses

We now consider how much of the explanatory power of V_f/P for long-term returns is due to its correlation with firm size and B/P. Fama and French (1992) show that both firm size and B/P have predictive power for cross-sectional returns. We examine the extent to which these two factors explain the predictive power of V_f/P.

Table 4
Average 36-month buy-and-hold returns for bi-dimensional portfolios

This table reports the average 36-month realized returns for bi-dimensional portfolios, formed using two ex ante firm characteristics simultaneously. On June 30 of year t, all sample firms are sorted (independently) into quintiles according to the variables designated in the panels. Book value (B) is book equity per share in calendar year $t-1$. Price (P) is the stock price at the end of June in year t. Analyst-based EBO value (V_f) is a fundamental value measure derived using I/B/E/S consensus analyst forecasts to proxy for future earnings. Table values represent mean buy-and-hold returns for each portfolio over the next 36 months. The number of observations is shown in parentheses. Results for All Firms represent unconditional means. The $Q5-Q1$ diff. results represent differences in mean returns between the $Q5$ and $Q1$ quintiles, controlling for quintile membership in other variable. The statistical significance of this difference is derived using a Monte Carlo simulation technique. Specifically, we form empirical reference distributions by randomly assigning eligible firms into quintiles each year while holding quintile membership in the other variable constant. ***, **, * signify that the observed difference between the $Q5$ and $Q1$ quintiles is significantly different from those of the reference distribution at the 1%, 5% and 10% levels, respectively (one-tailed). The sample period is 1977–1991 (t = 77 to 91).

Panel A – Average 36-month buy-and-hold returns for portfolios formed on the basis of both firm size (in-sample quintiles) and V_f/P

		Analyst-based *EBO* value-to-market (V_f/P)						
		$Q1$ (Low V_f/P)	$Q2$	$Q3$	$Q4$	$Q5$ (High V_f/P)	All Firms	$Q5-Q1$ Diff.
Size quintiles	$Q1$ (Small ME)	0.319	0.383	0.418	0.525	0.590	0.459	0.271***
		497	(378)	(405)	(425)	(668)	(2373)	
	$Q2$	0.287	0.520	0.539	0.504	0.577	0.486	0.290***
		(480)	(500)	(519)	(447)	(460)	(2406)	
	$Q3$	0.383	0.427	0.486	0.557	0.653	0.489	0.270***
		(559)	(518)	(447)	(360)	(2392)	(508)	
	$Q4$	0.318	0.473	0.498	0.565	0.706	0.503	0.388***
		(529)	(535)	(511)	(554)	(407)	(2536)	
	$Q5$ (Large ME)	0.350	0.433	0.496	0.578	0.679	0.525	0.329***
		(370)	(498)	(542)	(665)	(595)	(2670)	
	All firms	0.331	0.450	0.491	0.549	0.637	0.493	
		(2384)	(2470)	(2495)	(2538)	(2490)	(12377)	
	$Q5-Q1$ diff.	0.031	0.050	0.078*	0.053*	0.089*		

Panel B – Average 36-month buy-and-hold returns for portfolios formed on the basis of both B/P and V_f/P

		Analyst-based *EBO* value-to-market (V_f/P)						
		$Q1$ (Low V_f/P)	$Q2$	$Q3$	$Q4$	$Q5$ (High V_f/P)	All Firms	$Q5-Q1$ Diff.
Book-to-market	$Q1$ (Low B/P)	0.316	0.468	0.342	0.457	0.634	0.407	0.318***
		(998)	(592)	(296)	(264)	(269)	(2419)	
	$Q2$	0.366	0.461	0.489	0.415	0.516	0.450	0.150***
		(495)	(694)	(573)	(333)	(344)	(2439)	
	$Q3$	0.396	0.440	0.530	0.576	0.566	0.513	0.170*
		(295)	(515)	(660)	(565)	(453)	(2488)	
	$Q4$	0.350	0.422	0.484	0.589	0.630	0.535	0.280***
		(210)	(341)	(533)	(866)	(600)	(2550)	
	$Q5$ (high B/P)	0.263	0.442	0.544	0.588	0.732	0.558	0.469***
		(386)	(328)	(433)	(510)	(824)	(2481)	
	All Firms	0.331	0.450	0.491	0.549	0.637	0.493	
		(2384)	(2470)	(2495)	(2538)	(2490)	(12377)	
	$Q5-Q1$ Diff.	−0.053	−0.026	0.202***	0.131	0.098		

To address this question, we examine future returns to V_f/P portfolios while controlling for ME and B/P. Table 4 reports the average realized return to 36-month buy-and-hold strategies for bi-dimensional portfolios. To construct this table, we independently sort firms into quintiles based on each partitioning variable as of the end of June each year. Stocks are then assigned to one of 25 portfolios based on their bi-dimensional ranking. Panel A reports portfolio returns for V_f/P and firm size (using in-sample size cutoffs).[17] Panel B reports portfolio returns for V_f/P and B/P.

We again assess the statistical significance of the difference between $Q1$ and $Q5$ portfolios using a Monte Carlo simulation technique. In this analysis, we hold quintile membership in the other variable constant while randomizing across the variable of interest. For example, to create the empirical distribution for V_f/P in Panel B, we assigned the total population of firms within each B/P quintile into random V_f/P quintiles each year (without replacement). This procedure controls for B/P membership each year as well as any serial correlation in year-to-year returns. The resulting empirical distribution allows us to assess the incremental usefulness of one variable in predicting returns after controlling for the other.[18]

Panel A shows that V_f/P has strong predictive power in all five size quintiles. The difference in $Q5$–$Q1$ returns range from 27.0% to 38.8%. The right column shows that these differences are statistically significant at the 1% level in each of five size quintiles. The bottom row shows that when firms are divided using in-sample size cutoffs, large firms slightly outperform small firms after controlling for V_f/P.

Panel B shows the interaction of the B/P and V_f/P effects. Looking down each column, we see that the B/P effect is much weaker, and no longer monotonic after controlling for V_f/P. Conversely, looking across each row, we observe a largely monotonic and statistically significant relation between V_f/P and returns. The simulation results show V_f/P explains long-run returns within all five B/P portfolios; however, the B/P effect survives in only one V_f/P quintile. Taken together, Panels A and B suggest that in longer time horizons, the predictive power of V_f/P for future returns is not explained by either B/P or firm size. Later, we examine the relative contribution of each variable in returns prediction using a multiple regression approach.

[17] Results are similar when we use NYSE-size quintiles, however some cells have very few observations.

[18] When a firm drops out over our holding period, a terminating return to the delisting date is computed. The proceeds from termination, if any, are equally assigned to surviving firms in the same portfolio. Firms that switch exchanges are traced to their new exchange listings and retained in their original portfolios.

5.4. The relation between analyst forecast errors and ex ante firm characteristics

In this part of the paper, we investigate the quality of I/B/E/S consensus forecasts. This investigation has two motivations. First, the reliability of V_f depends critically on the quality of the earnings forecast used. In using I/B/E/S consensus forecasts to estimate V_f, we implicitly assume that these forecasts are unbiased with respect to public information. If this assumption is not true, we should be able to further improve the ability of V_f/P to predict returns by incorporating the predictable errors. Second, the mispricing hypothesis suggests a relation between certain characteristics and the *direction* of subsequent forecast errors, while the risk hypothesis does not. Therefore, this investigation should be helpful in distinguishing between the two hypotheses.[19]

Specifically, we investigate the relation between analyst forecast errors and four ex ante firm characteristics – the book-to-market ratio (B/P), past sales growth (SG), analyst consensus long-term earnings growth forecast (Ltg), and a new measure we call OP (for analyst optimism). The use of SG is suggested by Lakonishok et al. (1994)'s [LSV] finding that firms with higher (lower) past sales growth earn lower (higher) subsequent returns. Like LSV, we define SG in terms of the percentage growth in sales over the past five years. LSV argue that their finding is due to investor over optimism (pessimism) in firms with high (low) past sales growth. Thus, the mispricing hypothesis predicts that high (low) SGs are associated with over optimistic (pessimistic) I/B/E/S forecasts.

The use of the consensus long-term earnings growth forecast (Ltg) is motivated by LaPorta (1996) and Dechow and Sloan (1997). LaPorta shows that firms with higher long-term earnings forecasts (high Ltg firms) tend to earn lower subsequent returns. Dechow and Sloan (1997) show that the Ltg effect accounts for a significant portion of the return to contrarian investment strategies, including strategies based on the B/P and E/P ratios. We extend this literature by examining the power of Ltg to predict errors in long-term analyst forecasts, alone and in combination with variables.[20]

OP is a measure of analyst optimism derived from EBO fundamental value measures. Specifically, $OP = (V_f - V_h)/|V_h|$. OP measures the extent to which equity values based on analyst forecasts deviate from similar valuations based on historical earnings. Past studies show that analysts are more accurate than

[19] Similar arguments have been made in Dechow and Sloan (1995), Daniel and Mande (1994), LaPorta (1996), LaPorta et al. (1997). Unique features of our study include the use of a comprehensive valuation model that utilizes earnings forecasts the development of a prediction model for analyst forecast errors, as well as a trading strategy that directly exploits the predicted error in analysts.

[20] Ltg was not reported by I/B/E/S until 1981. In the pre-1981 period, we estimated the long-term growth rate using the growth rate implicit between $FY1$ and $FY2$.

earnings forecasts based on time-series models. We find the same is true for our sample of firms. However, in the cross-section, analyst forecasts that deviate the most from historical earnings may reflect an under-weighting of historical information. *OP* captures these deviations. Specifically, *OP* captures analyst optimism relative to past reported *ROE*s – analysts expect the highest (lowest) *OP* firms to experience the most (least) *ROE* growth. If analysts underweight historical profitability benchmarks in making their forecasts, then *OP* will be positively correlated with analyst optimism. Two examples from the behavioral literature that discuss this possibility are Tversky and Kahneman (1984) and DeBondt (1993).

Table 5 reports characteristics of quintile portfolios formed at the end of June each year by *B/P*, *SG*, *OP*, and *Ltg*. ROE_{t-1} is the actual reported returns-on-equity

Table 5
Accounting profitability of quintile portfolios formed by *B/P*, *SG*, *Ltg*, and *OP*

This table reports the accounting profitability characteristics of quintile portfolios formed at the end of June each year. *B/P* is the book-to-market ratio, where *B* is the book value per share for the fiscal year ended in year $t-1$, and *P* is the stock price at the end of June in year *t*. *SG* is the five-year growth rate in sales from period $t-6$ to $t-1$. *Ltg* is the consensus long-term earnings growth forecast from I/B/E/S as of May in year *t*. $OP = (V_f - V_h)/|V_h|$, where V_f is an *EBO* value derived using current I/B/E/S consensus forecasts and V_h is a similar *EBO* value measure derived using historical ROEs. *OP* measures the extent to which equity values based on analyst forecasts deviate from similar valuations based on historical earnings. Each panel reports mean values for individual quintiles. *Ret*36 is the average three-year buy-and-hold return for the portfolio. ROE_i is equal to reported net income in year *i* divided by the average of year *i* and $i-1$ book values. The forecast error for each observation ($FErr_{t+i}$) is computed by subtracting the forecasted ($FROE_{t+i}$) from the actual reported ROE in period $t+i$. To be included, firms are required to have all of the above variables available. Obs. is the number of observations. Results for All Firms represent unconditional means. $Q5-Q1$ Diff. results represent differences in means between the top (*Q*5) and bottom (*Q*1) quintiles. The statistical significance of this difference is derived using Monte Carlo simulation. Specifically, we form empirical reference distributions by randomly assigning eligible firms into quintiles each year. ***, **, * signify that the observed difference between the extreme quintiles is significantly different from those of the reference distribution at the 1%, 5% and 10% levels, respectively (one-tailed). Sample period is 1977-1991 ($t = 77$ to 91).

Panel A – Book-to-market (*B/P*) portfolios

	*Q*1 (Low *B/P*)	*Q*2	*Q*3	*Q*4	*Q*5 (High *B/P*)	All Firms	*Q*5 − *Q*1 Diff.
B/P	0.265	0.455	0.642	0.846	1.366	0.716	–
ROE_{t-1}	0.202	0.157	0.124	0.108	0.053	0.129	−0.149
$FROE_t$	0.285	0.200	0.158	0.131	0.080	0.170	−0.205
$FROE_{t+1}$	0.289	0.208	0.171	0.144	0.104	0.183	−0.185
$FROE_{t+2}$	0.280	0.207	0.173	0.147	0.107	0.183	−0.173
$FErr_t$	0.037	0.025	0.028	0.031	0.060	0.036	0.023***
$FErr_{t+1}$	0.054	0.058	0.062	0.051	0.073	0.059	0.019**
$FErr_{t+2}$	0.113	0.051	0.066	0.049	0.074	0.070	−0.039**
*Ret*36	0.462	0.530	0.543	0.617	0.613	0.553	0.151***
Obs.	2350	2378	2372	2375	2386	11861	

Panel B – Sales growth (*SG*) portfolios

	*Q*1 (Low *SG*)	*Q*2	*Q*3	*Q*4	*Q*5 (High *SG*)	All Firms	*Q*5 − *Q*1 Diff.
SG	− 0.014	0.451	0.787	1.322	9.081	2.331	–
ROE_{t-1}	0.087	0.125	0.143	0.148	0.141	0.129	0.054
$FROE_t$	0.132	0.161	0.176	0.185	0.199	0.170	0.067
$FROE_{t+1}$	0.154	0.174	0.186	0.192	0.207	0.183	0.053
$FROE_{t+2}$	0.155	0.174	0.185	0.192	0.207	0.183	0.052
FErrt	0.034	0.024	0.027	0.031	0.067	0.036	0.033***
$FErr_{t+1}$	0.060	0.051	0.046	0.051	0.092	0.059	0.032***
$FErr_{t+2}$	0.066	0.057	0.060	0.053	0.118	0.070	0.052**
*Ret*36	0.637	0.626	0.558	0.556	0.384	0.553	0.253***
Obs.	2365	2372	2372	2372	2380	1 1861	

Panel C – Analyst optimism (*OP*) portfolios

	*Q*1 (Low *OP*)	*Q*2	*Q*3	*Q*4	*Q*5 (High *OP*)	All Firms	*Q*5 − *Q*1 Diff.
OP	− 0.018	0.125	0.283	0.646	5.23	1.225	–
ROE_{t-1}	0.179	0.166	0.162	0.120	0.017	0.129	− 0.162
$FROE_t$	0.152	0.179	0.189	0.176	0.156	0.170	0.002
$FROE_{t+1}$	0.148	0.178	0.194	0.192	0.201	0.183	0.053
$FROE_{t+2}$	0.147	0.177	0.194	0.193	0.202	0.183	0.055
$FErr_t$	0.025	0.016	0.022	0.038	0.081	0.036	0.056***
$FErr_{t+1}$	0.050	0.030	0.041	0.052	0.126	0.059	0.076***
$FErr_{t+2}$	0.062	0.059	0.046	0.074	0.114	0.070	0.052***
*Ret*36	0.554	0.643	0.591	0.521	0.455	0.553	− 0.099***
Obs.	2365	2372	2372	2372	2380	1 1861	

Panel D – Long-term growth forecast (*Ltg*) portfolios

	Q1 (Low *Ltg*)	*Q*2	*Q*3	*Q*4	*Q*5 (High *Ltg*)	All Firms	*Q*5 − *Q*1 Diff.
Ltg	0.046	0.105	0.135	0.169	0.306	0.153	–
ROE_{t-1}	0.105	0.122	0.145	0.147	0.126	0.129	0.021
$FROE_t$	0.135	0.155	0.179	0.193	0.189	0.170	0.054
$FROE_{t+1}$	0.140	0.166	0.191	0.205	0.212	0.183	0.072
$FROE_{t+2}$	0.139	0.165	0.190	0.202	0.215	0.183	0.076
$FErr_t$	0.028	0.034	0.026	0.032	0.062	0.036	0.034***
$FErr_{t+1}$	0.042	0.055	0.057	0.051	0.093	0.059	0.051***
$FErr_{t+2}$	0.042	0.045	0.083	0.063	0.122	0.070	0.080***
*Ret*36	0.636	0.610	0.560	0.539	0.426	0.553	0.210***
Obs.	2336	2365	2373	2389	2398	1 1861	

for years $t-1$. $FROE_{t+i}$ is the predicted ROE for year $t+i$ based on I/B/E/S analyst forecasts, and $FErr_{t+i}$ is the average forecast error in the year $t+i$ forecasts. Specifically, $FErr_{t+i}$ is computed by subtracting each firm's predicted year $t+i$ ROE ($FROE_{t+i}$) from the actual reported ROE in period $t+i$ (ROE_{t+i}), and averaging across all firms. Analyst over-optimism (pessimism) relative to future reported earnings results in more positive (negative) $FErr$ values.[21] $Ret36$ is the average three-year buy-and-hold return for each portfolio. Statistical significance of the difference in means between extreme quintiles is determined using Monte Carlo techniques.

Table 5 confirms several prior findings, while highlighting the importance of analyst forecast errors. Firms are included only if they have the five years of historical sales data necessary to compute SG. Consistent with prior studies (e.g., FF, 1995; Fairfield, 1994; Bernard, 1994), Panel A shows that lower (higher) B/P firms have higher (lower) reported $ROEs$. Both current year $ROEs$ (ROE_{t-1}) and three-year ahead $ROEs$ (ROE_{t+2}) are significantly higher for low B/P firms. Analysts are also predicting higher profitability for higher P/B firms: the average $FROE_{t+2}$ is higher (lower) for low (high) B/P firms. However, a proper investigation of market efficiency should focus, not on forecasted or actual $ROEs$, but on the pattern of *errors* in forecasted $ROEs$ ($FErr_{t+i}$).

Consistent with prior studies – e.g., Fried and Givoly (1982), O'Brien (1988) – Table 5 shows that analysts are, on average, overly-optimistic: $FErr_{t+i}$ is positive for all quintiles in all three Panels. The magnitude of the bias in one-year-ahead forecasts is comparable to those reported in prior studies (e.g., O'Brien, 1988, Table 3). The two- and three-year-ahead biases are somewhat higher, reflecting the compounding effects of pevious year forecast errors, as well as our use of the long-term growth rate to estimate three-year-ahead $FROEs$.

More importantly, this table reveals several interesting cross-sectional patterns in analyst forecast errors. Panel A shows a negative relation between B/P and $FErr_{t+2}$ – analysts are most over-optimistic in low B/P firms. However, this relation is not monotonic across the quintiles, and does not hold in one- and two-year-ahead forecasts. This result suggests that the B/P effect is only tangentially related to $FErr$.

Panel B shows that SG is positively correlated with $FErr_{t+2}$. While this finding is consistent with the LSV mispricing conjecture, the effect is again not monotonic. Analyst over-optimism increases sharply for the highest SG quintile, but is otherwise flat. Similarly, buy-and-hold returns are flat for quintiles $Q1$

[21] We selected this definition so that analyst over-optimism results in positive forecast errors. In most prior studies, forecast errors are defined with the opposite sign (e.g., O'Brien, 1988). Note also that we report the forecast errors for the next three years. In subsequent tests, we focus on forecast errors in three-year ahead ROE forecasts because the longer-term ROEs have the greatest impact on our V_f estimate. The use of ROE rather than earnings-per-share (EPS) mitigates stock split timing problems encountered when comparing forecasted and actual EPSs.

through $Q4$, but drop sharply for the top SG quintile. This result is consistent with Dechow and Sloan (1997), who also find that firm rankings on past growth measures (both earnings and sales) do not result in a strong systematic return differentials in intermediate portfolios. Overall, the evidence suggests SG is related to analyst forecast errors in the direction predicted by the mispricing hypothesis, but the relation is non-linear.

Panel C shows that the OP portfolios are also related to analyst forecast errors. As predicted by the mispricing hypothesis, analyst forecasts tend to be more over-optimistic for high OP firms in all forecast horizons. Similarly, Panel D shows that analyst forecasts tend to be more over-optimistic for high Ltg firms. Panel D shows that Ltg has strong predictive power for returns, and unlike SG, the intermediate portfolio returns to Ltg rankings are monotonic. Taken together, Panels A through D show that all four variables appear to have some predictive power for cross-sectional differences in long-term analyst forecast errors.

To evaluate the robustness of these relations over time, Table 6 reports the result of 15 annual cross-sectional regressions of realized analyst forecast errors on each of these four firm characteristics (year $t = 1977$–1991). The dependent variable for each regression is $FErr_{t+2}$. The independent variables are SG, B/P, OP, or Ltg. To facilitate interpretation of these results and to reduce the effect of outliers, the independent variables are expressed in terms of their percentile ranks. To compute its percentile rank, each variable is sorted as of the end of June in year t and assigned to percentiles.

Table 6 shows that the relation between these four variables and the subsequent analyst forecast error is robust over time. Model 1 shows that low (high) B/Ps are generally associated with ex post analyst optimism (pessimism). Model 2 shows that high (low) past sales growth (SG) is positively associated with excessive optimism (pessimism). Model 3 shows that when V_{f} is much higher (lower) than V_{h}, analysts tend to be too optimistic (pessimistic). Finally, Model 4 shows that high Ltg firms tend to have overly optimistic forecasts. The sign of the estimated coefficient is correct in 55 out of 60 individual cases. Newey–West (1987) t-statistics based on time-series variations in the annual estimates (bottom row) indicate statistical significant at the 1% level for all four models.

Because the independent variables are all expressed in terms of percentile ranks, we can compare their estimated coefficients. Based on the last row in Table 6, the top B/P ranked percentile firms have forecasted ROE errors that are 2.5% lower than those of the bottom percentile B/P firms. Similarly, the difference in forecasted ROE errors for the top and bottom SG percentiles is 4.3%. The difference between top and bottom percentile OP firms is 7.0%, while the difference between top and bottom percentile Ltg firms is 7.3%. Since the typical firm earns an ROE of 13%, these differences appear substantial and economically significant.

Table 6
The relation between analyst forecast errors and ex ante firm characteristics

This table reports the results of 15 (t = 1976–1990) cross-sectional regressions of realized analyst forecast errors on four firm characteristics. Forecast errors (the dependent variable) are computed by subtracting actual returns-on-equity (ROEs) in period $t+2$ from predicted $t+2$ ROEs obtained from I/B/E/S consensus earning forecasts available prior to June 30 of year t. The independent variables are computed as follows. SG is the five-year percentage growth in sales from period $t-6$ to $t-1$. B/P is the book-to-market ratio, where B is the reported book value per share for the fiscal year ended in year $t-1$ and P is the stock price at the end of June in year t. OP is a measure of analyst optimism derived using Edward–Bell–Ohlson (EBO) fundamental value measures. Specifically, $OP = (V_f - V_h)/|V_h|$, where V_f is an EBO value derived using current I/B/E/S analyst consensus forecasts and V_h is a similar EBO value measure derived using historical ROEs. Ltg is the consensus long-term earnings growth forecast from I/B/E/S as of May of year t. $RK(.)$ is a percentile rank operator. To compute percentile ranks, the independent variables are sorted as of the end of June each year and assigned to percentiles. Numbers in the All Years row represent time-series means and Newey–West (1987) t-statistics based on time-series variation in the annual estimates. ***, **, and * signify statistical significance at the 1%, 5% and 10% levels respectively.

		Model 1	Model 2	Model 3	Model 4	
	Year t	$RK(B/P)$	$RK(SG)$	$RK(OP)$	$RK(Ltg)$	*Obs.*
Coeff.	1976	– 0.020	0.011	0.049	0.049	206
t-stat		– 1.20	0.67	3.01***	3.02***	
Coeff.	1977	0.005	0.016	0.035	0.017	173
t-stat		0.25	0.77	1.71**	0.83	
Coeff.	1978	– 0.007	0.001	– 0.016	– 0.023	261
t-stat		– 0.45	0.64	– 0.92	– 1.27	
Coeff.	1979	– 0.009	0.029	0.015	0.022	387
t-stat		– 0.68	2.12***	1.10	1.56*	
Coeff.	1980	– 0.050	0.036	0.061	0.081	631
t-stat		– 3.28***	2.37***	4.02***	5.40***	
Coeff.	1981	– 0.130	0.088	0.140	0.150	684
t-stat		– 8.09***	5.27***	8.71***	9.48***	
Coeff.	1982	– 0.013	0.060	0.003	0.115	694
t-stat		– 0.82	3.87***	0.20	7.66***	
Coeff.	1983	– 0.074	0.043	0.143	0.128	684
t-stat		– 4.00***	2.25***	7.97***	7.11***	
Coeff.	1984	– 0.029	0.046	0.169	0.086	729
t-stat		– 1.59*	2.56***	11.19***	4.96***	
Coeff.	1985	– 0.000	0.048	0.068	0.100	691
t-stat		– 0.016	2.93***	4.24***	6.40***	
Coeff.	1986	0.002	0.067	0.085	0.081	669
t-stat		0.104	3.96***	5.13***	4.89***	
Coeff.	1987	– 0.033	0.040	0.113	0.090	678
t-stat		– 1.84**	2.19***	6.50***	5.09***	
Coeff.	1988	– 0.008	0.077	0.069	0.064	678
t-stat		– 0.447	4.37***	4.03***	3.68***	
Coeff.	1989	0.001	0.047	0.029	0.059	756
t-stat		0.028	2.47***	1.54*	3.19***	
Coeff.	1990	– 0.010	0.036	0.081	0.078	779
t-stat		– 0.647	2.20**	5.09***	4.92***	
Mean	All	– 0.025	0.043	0.070	0.073	15 years
t-stat	Years	– 2.69***	7.00***	5.01***	6.57***	

Table 7
Predicting analyst forecast errors using multiple firm characteristics

This table reports the results of 15 (t = 1976–1990) multiple cross-sectional regressions of realized analyst forecast errors on four firm characteristics. Forecast errors for each firm (the dependent variable) are computed by subtracting actual returns-on-equity (ROEs) in period $t+2$ from predicted $t+2$ ROEs obtained from I/B/E/S consensus earning forecasts. To reduce the effect of outliers, the top and bottom 1% forecasted error each year are omitted. The independent variables are computed as follows. SG is five year percentage growth in sales from period $t-6$ to $t-1$. B/P is the book-to-market ratio, where B is the reported book value per share for the fiscal year ended in year $t-1$ and P is the stock price at the end of June in year t. $OP = (V_f - V_h)/|V_h|$; where V_f is an EBO value derived using current I/B/E/S analysts consensus forecasts and V_h is a similar EBO value measure derived using historical ROEs. Ltg is the consensus long-term earnings growth forecast from I/B/E/S. $RK(.)$ is a percentile rank operator. To compute percentile ranks, the independent variables are sorted as of the end of June each year, and assigned to percentiles. Numbers in the All Years row represent time-series means and Newey–West (1987) t-statistics based on time-series variation in the annual estimates. ***, ** and * signify one-tailed statistical significance at the 1%, 5% and 10% levels, respectively.

	Year t	$RK(SG)$	$RK(B/P)$	$RK(OP)$	$RK(Ltg)$	R^2	F-statistic	$Obs.$
Coeff.	1977	0.010	− 0.013	0.032	0.031	0.061	3.31**	206
t-stat		0.58	− 0.74	1.62	1.62			
coeff.	1978	0.020	0.015	0.042	− 0.010	0.023	1.00	173
t-stat		0.92	0.66	1.52	− 0.35			
Coeff.	1979	− 0.002	− 0.019	− 0.012	− 0.020	0.010	0.62	261
t-stat		− 0.09	− 0.89	− 0.49	− 0.91			
Coeff.	1980	0.029	0.009	0.006	0.018	0.017	1.64	387
t-stat		2.01	0.57	0.37	1.03			
Coeff.	1981	0.014	− 0.020	0.023	0.065	0.055	9.13***	631
t-stat		0.88	− 1.13	1.26	3.98			
Coeff.	1982	0.040	− 0.073	0.076	0.096	0.195	41.2***	684
t-stat		2.36	− 4.25	4.43	5.64			
Coeff.	1983	0.031	0.042	− 0.030	0.132	0.097	18.5***	694
t-stat		1.86	2.48	− 1.87	7.59			
Coeff.	1984	0.048	0.033	0.119	0.089	0.117	22.5***	684
t-stat		2.47	1.36	5.92	3.56			
Coeff.	1985	0.079	0.002	0.183	0.000	0.150	31.9***	729
t-stat		4.46	0.10	9.62	0.00			
Coeff.	1986	0.042	0.067	0.042	0.109	0.087	16.4***	691
t-stat		2.52	3.70	2.51	5.73			
Coeff.	1987	0.062	0.044	0.068	0.055	0.075	13.5***	669
t-stat		3.46	2.44	3.82	2.71			
Coeff.	1988	0.020	0.007	0.096	0.056	0.075	13.7***	678
t-stat		1.03	0.35	5.23	2.52			
Coeff.	1989	0.064	0.023	0.056	0.025	0.049	8.66***	678
t-stat		3.33	1.22	3.06	1.17			
Coeff.	1990	0.030	0.029	0.012	0.056	0.019	3.66***	756
t-stat		1.46	1.45	0.62	2.43			
Coeff.	1991	0.014	0.041	0.065	0.072	0.054	11.0***	779
t-stat		0.80	2.24	3.82	3.56			
Mean	All	0.035	0.010	0.051	0.050	0.074		15 years
t-stat	Years	5.94***	1.16	3.55***	4.26***			

5.5. Predictability of analyst forecast errors

We now assess the amount of cross-sectional variation in forecast errors that is explained by combining our four variables. Table 7 reports the results of annual multiple regressions of these forecast errors on ranked percentiles of B/P, SG, OP, and Ltg. With each variable conditioned on the others, OP, SG and Ltg provide greater explanatory power for forecast errors than B/P. In fact, the Newey-West (1987) time-series t-statistics in the last row show that after controlling for the other variables, B/P offers little incremental contribution to the model.

The R^2 for the annual regressions ranges from a low of 1% in 1979 to a high of 19.5% in 1982. The average R^2 is 7.4%, and the annual F-statistics indicate significance in 12 out of 15 years. Clearly, a modest, but consistent, portion of the error in the consensus I/B/E/S forecast is predictable each year. Although the predictable portion is not large, the consistency of the coefficients suggests a potential trading strategy. We examine this possibility in the next section.

5.6. Profiting from the forecast error

To exploit the predictable component of the I/B/E/S forecast error, we first estimate annual cross-sectional regressions of the form presented in Table 7. Specifically, we regress forecast errors realized in year $t-1$ on percentile ranks of SG, B/P, OP and Ltg from year $t-4$. From these annual regressions, we derive estimated coefficient weights for each variable. We then apply these estimated coefficients to each firm's period $t-1$ B/P, SG, OP, and Ltg variables to compute a predicted forecast error. Specifically, the predicted forecast error for firm i in portfolio formation year t is:

$$PE_{rr_{it}} = \hat{\alpha} + \hat{\beta}_1 RK(SG_{it-1}) + \hat{\beta}_2 RK(BP_{it-1}) + \hat{\beta}_3 RK(OP_{it-1}) + \hat{\beta}_4 RK(Ltg_{it-1}). \tag{4}$$

The parameters $\hat{\alpha}$, $\hat{\beta}_1$, $\hat{\beta}_2$, $\hat{\beta}_3$ and $\hat{\beta}_4$ are estimated from rolling cross-sectional regressions based on year $t-4$ information and actual year $t-1$ earnings. $RK(.)$ is the percentile rank operator. Large positive (negative) values of $PErr$ correspond to excessively optimism (pessimism) forecasts. Since three past years of data are necessary to estimate $PErr$, the sample period for this test is 1979–1992.

Table 8 presents returns to a $PErr$ strategy and compares the results to returns from other investment strategies. To construct this table, we regressed the one- and three-year-ahead buy-and-hold returns for our sample firm-years on scaled decile ranks of ME, B/P, V_f/P, and $PErr$. Following Bernard and Thomas (1990) and Dechow and Sloan (1997), we use scaled decile rankings for

all independent variables. For each calendar year, the independent variables are assigned in descending order to deciles, and then scaled so that they range from zero (for the lowest decile) to one (for the highest decile). This approach allows the regression coefficients to be interpreted as estimates of the return to a zero investment portfolio with a long position in the stocks in the highest decile and a short position in the stocks of the lowest decile.

Table 8 shows that a decile-based *PErr* strategy yields approximately 4% over the next 12-months, and 27.7% over the next 36-months. Both results are statistically significant at the 1% level. Clearly, *PErr* has significant predictive power for one- and three-year-ahead returns. Over the next 12 months, the *PErr* strategy performs about as well as V_f/P and B/P. Over the next 36 months, the *PErr* strategy outperforms the B/P strategy, but underperforms the V_f/P strategy.

Models 5 and 6 in Table 8 evaluate the incremental contribution V_f/P and *PErr* controlling for B/P and ME. These results show that V_f/P has significant incremental power to predict cross-sectional returns in both one- and three-year-ahead regressions. Model 5 in Panel B shows that over 36 months, V_f/P is the most important variable in explaining cross-sectional returns. This evidence extends the results in Table 4 by illustrating that V_f/P has incremental predictive power controlling for both ME and B/P. In addition, Model 6 in both panels shows that the *PErr* strategy enhances the predictive power of V_f/P, even controlling for ME and B/P.

Fig. 2 presents a comparison of the cumulative monthly returns produced by four alternative trading strategies: a B/P strategy, a *PErr* strategy, a V_f/P strategy, and a combined strategy. Returns to the B/P and V_f/P strategies are based on buying firms in the top quintile and selling firms in the bottom quintile each year. The results are the same as those reported in Fig. 1. For the *PErr* strategy, cumulative returns are the average returns from selling firms in the top quintile (high *PErr* firms) and buying firms in the bottom quintile (low *PErr* firms). For the combined strategy, we buy (sell) firms that are simultaneously in the top (bottom) V_f/P quintile and the bottom (top) *PErr* quintile. Fig. 2 shows that the highest returns are produced by the combined strategy, indicating that the *PErr*-based strategy has incremental explanatory power to V_f/P. Indeed, over a 36-month holding period, this combined strategy yields a cumulative return of 45.5%.

Table 9 reports the year-by-year results of implementing each strategy. This test has less statistical power to detect abnormal returns, but provides a better picture of the robustness of each strategy over time. The results show that none of the strategies perform particularly well over one-year holding periods. However, over three-year holding periods, *PErr*, V_f/P and the combined strategy all outperform the B/P strategy.

In the early years (pre-1982), the V_f/P strategy does better than the combined strategy. Recall from Table 7 that during these years our prediction model

Table 8
Relative importance of ex ante firm characteristics in returns prediction

This table presents estimated coefficients from regressions of one- and three-year-ahead stock returns on various ex ante firm characteristics. The sample consists of 11,861 firm-years between 1976 and 1992 for firms with all data available. Dependent variables are the one- and three-year-ahead buy-and-hold returns, including dividends and any liquidating distributions. The return cumulation period begins at the end of June in each year t. The independent variables are market value of equity (ME), book-to-price (BP), value-to-price (V_f/P), and the predicted analyst forecast error ($PErr$). Market value of equity (ME) and stock price (P) are as of June 30 of each year t. Book value (B) is book equity per share in calendar year $t-1$. V_f is a fundamental value estimate derived using current I/B/E/S consensus forecasts available prior to June 30 of year t. $PErr$ is the predicted error in the year t consensus forecast of year $t+2$ ROEs. For firm i and period t, the predicted forecast error is

$$PErr_{it} = \hat{\alpha} + \hat{\beta}_1 RK(SG_{it-1}) + \hat{\beta}_2 RK(BP_{it-1}) + \hat{\beta}_3 RK(OP_{it-1}) + \hat{\beta}_4 RK(Ltg_{it-1}).$$

This predicted error is estimated using information available prior to June of year t. Specifically, we require each firm's five-year past sales growth (SG), market-to-book ratio (BP), long-term consensus earnings growth forecast (Ltg), and an optimism measure ($OP = (V_f - V_h)/|V_h|$), where V_h is similar to V_f, but derived using historical earnings rather than analyst forecasts. $RK(.)$ is a percentile rank operator. The parameters $\hat{\alpha}$, $\hat{\beta}_1$ $\hat{\beta}_2$ $\hat{\beta}_3$, and $\hat{\beta}_4$ are estimated from rolling cross-sectional regressions based on year $t-4$ information and year $t-1$ reported earnings. Large positive (negative) values of $PErr$ correspond to predictions of excessive over-optimism (pessimism). All independent variables are assigned in descending order to deciles, and then scaled so that they range from zero (for the lowest decile) to one (for the highest decile). ***, **, * Denote significance at the 1%, 5% and 10% levels, respectively, using a two-tailed t-test.

Panel A: One-year-ahead returns

Model	Intercept	BP	ME	V_f/P	$PErr$	Adj. R^2 (%)
1	0.151***	0.051***	–	–	–	0.12
2	0.186***	–	– 0.019	–	–	0.01
3	0.155***	–	–	0.042***	–	0.09
4	0.196***	–	–	–	– 0.040***	0.07
5	0.147***	0.039***	– 0.013	0.031**	–	0.15
6	0.176***	0.029*	– 0.023	0.030**	– 0.035**	0.19
Panel B: Three-year-ahead returns						
1	0.468***	0.168***	–	–	–	0.37
2	0.538***	–	0.026	–	–	0.00
3	0.365***	–	–	0.370***	–	1.83
4	0.688***	–	–	–	– 0.277***	1.03
5	0.341***	0.051	0.013	0.352***	–	1.83
6	0.539***	– 0.029	– 0.053*	0.343***	– 0.241***	2.47

performed poorly. Thus, the PErr strategy appears to add noise without adding additional predictive power. However, since 1982, the combined strategy outperformed V_f/P every year. The 36-month return to the combined strategy is consistently positive through up and down markets. Yet the strategy is not

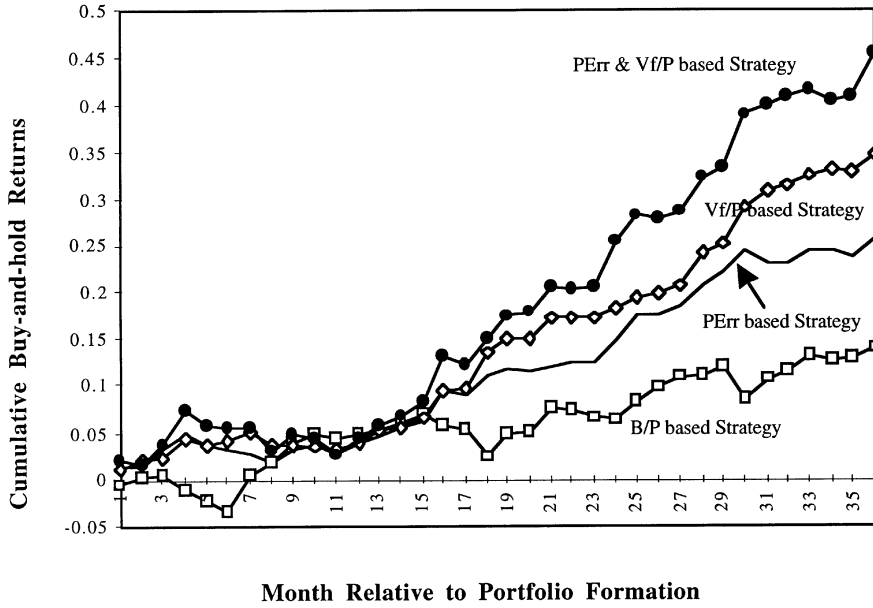

Fig. 2. A comparison of cumulative (buy-and-hold) returns from alternative trading strategies. This figure shows the cumulative returns from several alternative trading strategies. B is book equity per share in calendar year $t-1$. P is price per share at the end of June 30 of year t. PErr is the predicted error in consensus analyst forecasts for year $t+2$ return-on-equity (ROE), estimated as of June 30 of year t. V_f is a fundamental value estimate based on the consensus analyst forecast. Each year, portfolios are formed at the end of June by sorting firms into quintiles on the basis of B/P, $PErr$, and V_f/P. For the B/P and V_f/P based strategies, this graph depicts the cumulative buy-and-hold returns produced by buying firms in the top quintile and selling firms in the bottom quinitle at the beginning of July, and maintaining these investments until the end of the indicated month. The $PErr$-based strategy is simialr, except firms in the top quintile are sold and firms in the bottom quintile are purchased. For the combined $PErr$ and V_fP strategy, firms are included in the long (short) portfolio if they are simultaneously in the top V_f/P quintile and the bottom PErr quintile. The sample period is 1979–1991 (year $t = 1970$ to 1991).

without risk, particularly over a one-year horizon. In 1981, the combined strategy lost 66%, due to large losses in a few stocks.

6. Summary

In this study, we operationalized an analyst-based residual income model and used the resulting value-to-price (V_f/P) ratio to examine issues related to market efficiency and the predictability of cross-sectional stock returns. Our results show that V_f/P is a reliable predictor of cross-sectional returns, particularly over

Table 9
Year-by-year returns to various trading strategies

This table reports the cumulative one-year and three-year buy-and-hold returns from alternative trading strategies. B is book equity per share in calendar year $t-1$. P is price of the stock at the end of June 30 in year t. $PErr$ is the predicted error in consensus analyst forecasts of year $t+2$ return-on-equity. V_f is the EBO fundamental value measure based on the consensus analyst forecast. Each year, portfolios are formed at the end of June by sorting firms into quintiles on the basis of B/P, $PErr$, and V_f/P. For B/P- and V_f/P-based strategies (Panels A and C, respectively), the table values represent the average cumulative equal-weighted returns produced by buying firms in the top quintile and selling firms in the bottom quintile at the beginning of July of each year, and maintaining these investments for either 12 or 36 months. The $PErr$-based strategy (Panel B) is similar, except firms in the top quintile are sold and firms in the bottom quintile are bought. For the combined $PErr$ and V_f/P strategy (Panel D), firms are included in the long (short) portfolio if they are simultaneously in the top V_f/P quintile and the bottom $PErr$ quintile. Numbers in the Mean row represent time-series means of the annual returns. Reported t-statistics are based on time-series variations in the annual means, with Newey–West (1987) correction for serial correlation. Number of firms indicates the highest and lowest number of trading positions taken per year, where a trading position maybe either a long position or a short position. ***, ** and * signify one-tailed statistical significance at the 1%, 5% and 10% levels respectively.

	Panel A		Panel B		Panel C		Panel D	
	B/P (high–low)		$PErr$ (low–high)		V_f/P (high–low)		Combined $PErr$ and V_f/P	
Year t	1-year	3-year	1-year	3-year	1-year	3-year	1-year	3-year
78	−0.159	0.043	−0.247	−0.294	−0.102	0.325	−0.239	0.171
79	0.091	0.675	0.062	0.338	0.052	0.553	0.022	0.456
80	0.244	0.431	−0.155	−0.092	0.244	0.447	0.077	0.295
81	−0.158	0.286	−0.322	0.529	−0.130	0.393	−0.659	0.909
82	0.234	0.670	0.273	0.745	0.274	1.127	0.343	1.204
83	0.118	0.199	0.261	0.349	0.25	0.589	0.363	0.411
84	−0.105	0.044	0.129	0.264	0.075	−0.002	0.133	0.171
85	0.063	0.176	0.071	0.397	−0.083	0.017	−0.104	0.328
86	0.05	0.052	0.103	0.183	0.072	0.141	0.113	0.228
87	0.199	0.083	0.124	0.221	0.057	0.250	0.190	0.410
88	−0.176	−0.110	−0.019	0.188	−0.062	0.136	−0.038	0.499
89	0.008	0.189	0.074	0.322	0.099	0.214	0.109	0.402
90	0.091	–	0.066	–	−0.027	–	0.005	–
91	0.147	–	−0.038	–	−0.184	–	−0.090	–
Mean	0.046	0.228	0.027	0.263	0.038	0.349	0.016	0.457
t-stat	1.23	3.32**	0.62	3.55***	1.02	4.06***	0.07	5.40***
Number of firms	164 to 442	164 to 392	164 to 442	164 to 393	164 to 442	164 to 393	48 to 149	48 to 149

longer horizons. Over the next 12-months, the predictive power of V_f/P is comparable to that of B/P. However, over the next 36-months, V_f/P has much stronger predictive power than B/P. This ability to predict long-term returns is not attributable to B/P, firm size, or beta.

Because of its importance in estimating V_f, we also investigate the reliability of long-term I/B/E/S consensus earnings forecasts. We find that cross-sectional errors in three-year-ahead consensus forecast are predictable. Specifically, we find some evidence that analysts tend to be more overly-optimistic in firms with higher past sales growth and P/B ratios. In addition, we find stronger evidence of over-optimism in firms with higher forecasted earnings growth (*Ltg*) and higher forecasted *ROEs* relative to current *ROEs* (*OP*). Combining these variables in a prediction model, we develop an estimate of the prediction error in long-term forecasts (*PErr*), and show this estimate has predictive power for cross-sectional returns. Moreover, we show this predictive power is incremental to a V_f/P strategy, and a combined strategy yields the highest returns.

Our evidence suggests that firm value estimates based on a residual income model may be a useful starting point for predicting cross-sectional stock returns. Much recent research has focused accounting-based ratios that exhibit predictive power for stock returns. The B/P ratio, in particular, has received significant attention. Our results suggest that superior return prediction may result from adopting a more complete valuation approach.

Our implementation of the residual income model is simple, and leaves much room for improvement. While we focus on an analyst-based valuation model, future work may focus on alternative mechanical models of earnings prediction (e.g., Dechow et al., 1997). Future studies may also lead to refinements in other key parameters of the model, including forecasted dividend payout ratios and cross-sectional variations in discount rates. We hope that our findings will encourage further research along these lines.

Our finding that prices converge to value estimates gradually over longer horizons (beyond 12-months) is puzzling. The effect may be due, in part, to the conservative nature of our tests. In a recent replication of our results, Herzberg (1998) shows a strong partial price correction in the first month after the strategy is implementable (see Exhibit 12). However, we implement this strategy with a 5 to 6 week lag – we use IBES forecasts publicly available by the third week of May and form portfolios as of June 30th. Our first-year results therefore do not capture the short-term profits from the first 6 weeks.

This explanation suggests our first year returns should be higher, but it does not explain why returns remain high in years two and three. One explanation is that the price convergence to value is a much slower process than prior evidence suggests. This possibility raises interesting questions about the efficiency of the market, and in particular, about the process by which information about long-term fundamentals is impounded in price. The bias in long-term analyst

forecasts is not realized until two or three years into the future, and analysts appear to revise their long-term forecasts only gradually over time. Our evidence on the predictability of long-term forecast errors in consensus forecasts is consistent with this long-term mispricing hypothesis. However, it is difficult to understand why arbitrage forces do not eliminate this pricing anomaly more quickly.

Alternatively, V_f/P may be yet another proxy for cross-sectional risk differences. Our tests control for two obvious sources of potential risk: the B/P ratio and firm size. We find that high V_f/P firms generally have lower market betas, so sensitivity to overall market movements is an unlikely explanation for their higher subsequent returns. In addition, unlike returns to a B/P strategy, returns to a V_f/P strategy exhibits a pattern of lower short-term returns and higher long-term returns. This pattern is difficult to reconcile with a risk explanation. Despite these concerns, we acknowledge that high V_f/P firms may still be riskier than low V_f/P firms in some other, as yet unidentified, dimension. We leave this question to future research.

7. For Further Reading

The following references are also of interest to the reader: Dechow et al. (1998) and Fama and French (1955).

Acknowledgements

We thank the late Victor Bernard, whose many insights helped to bring the residual income valuation model to life. We also thank Jeff Abarbanell, Jim Bodurtha, Larry Brown, John Core, Kent Daniel, Tom Dyckman, Ken French, S.P. Kothari (Editor), Bruce Lehman, Pat O'Brien, Jay Shanken (the referee), Richard Sloan, Bhaskaran Swaminathan, an anonymous referee, and workshop participants at Cornell University, Dartmouth College, Georgetown University, Harvard University, the University of Minnesota, Ohio State University, the University of Oregon, the University of Rochester, and Yale University for helpful suggestions. James Myers provided expert research assistance. Steve Merritt, a Michigan MBA student, deserves credit for first identifying an apparent market anomaly with this trading rule during a class exercise. Earnings forecasts used in this paper are provided by I/B/E/S. We gratefully acknowledge the financial support of the Q-Group and the KPMG Peat Marwick Foundation (Lee).

Appendix A. Using I/B/E/S forecasts to derive future ROE estimates

Our implementation of the Edwards–Bell–Ohlson (EBO) formula requires three future ROE forecasts [$FROE_t$, $FROE_{t+1}$ and $FROE_{t+2}$]. We derive these

future ROEs from I/B/E/S consensus EPS estimates. Since year-end book values are dependent on current year ROEs, we use a sequential process to estimate future ROEs. The steps in the process are listed below. Year t refers to the year of portfolio formation.

Step 1: Estimating $FROE_t$ and B_t. We require that all sample firms have a one-year-ahead I/B/E/S consensus EPS forecast $[FY1]$. Forecasted ROE for year t is then computed as the year t consensus forecast, divided by the average book value per share during year $t-1$. Use of the average, rather than year-end, book value reduces the chance of an extremely low denominator. We then use $FROE_t$ and the dividend payout ratio (k) to derive the ending book value for year t. Notationally, we have:

$$FROE_t = FY1/[(B_{t-1} + B_{t-2})/2],$$

$$B_t = B_{t-1}[1 + FROE_t(1-k)].$$

Step 2: Estimating $FROE_{t+1}$ and B_{t+1}. We also require that all sample firms have a two-year-ahead consensus forecast $[FY2]$. We then compute $FROE_{t+1}$ and B_{t+1} analogously:

$$FROE_{t+1} = FY2/[(B_t + B_t - 1)/2], B_{t+1} = B_t[1 + FROE_{t+1}(1-k)].$$

Step 3: Estimating $FROE_{t+2}$ and B_{t+2}. Where a long-term earnings growth estimate $[Ltg]$ is available, we compute $FROE_{t+2}$ and B_{t+2} as follows:

$$FROE_{t+2} = [FY2(1 + Ltg)]/[(B_{t+1} + B_t)/2],$$

$$B_{t+2} = B_{t+1}[1 + FROE_{t+2}(1-k)].$$

Where Ltg is not available, we use $FROE_{t+1}$ to proxy for $FROE_{t+2}$.

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